FeatureScalpel: Mechanistic Interpretability And Intervention Of Large Language Models via Sparse Autoencoder

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by

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CERTIFICATE

This is to certify that the report entitled **FeatureScalpel: Mechanistic Interpretability And Intervention Of Large Language Models via Sparse Autoencoder** submitted by **Adithyan R S** (KTU Reg No: B23CS2109), **Alona Biju** (KTU Reg No: B23CS2113), **Alvin Binu** (KTU Reg No: B23CS2114) and **Darsan M** (KTU Reg No: B23CS2125) to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Computer Science and Engineering (AI) is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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Declaration

We hereby declare that the project report entitled **FeatureScalpel: Mechanistic Interpretability And Intervention Of Large Language Models via Sparse Autoencoder**, submitted to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI), is a record of original work carried out by us at the Department of Computer Science and Engineering, Mar Baselios College of Engineering and Technology (Autonomous), Thiruvananthapuram. The experimental results reported here were produced by the system described in this report. Wherever the work of others has been used, it has been duly acknowledged and cited. This report has not been submitted, in whole or in part, to any other University or Institute for the award of any degree or diploma.

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Abstract

Language models often rank a correct answer below competing words even though the answer is within reach. Fixing this by changing weights is permanent and gets worse as edits pile up, and adding text to the prompt does not use what the model already computes. This project, **FeatureScalpel**, asks whether the fix can be **transient and weight-free**. During one forward pass the residual stream of GPT-2 Small is edited through the features of a pretrained **sparse autoencoder (SAE)**, and the edit disappears when the pass ends.

The report follows the work as it happened, mistakes included. Muting the target's own helpers corrected 1 of 22 test facts and muting the leading competitor corrected 3. Muting alone failed on our probe prompt, while muting together with boosting brought it to rank 1. Safety filtering, a study of depth, a batched implementation (186 model calls per prompt and layer down to 6), pool refill and candidates from all prompt positions followed. A small network that chose the strengths did no better per prompt than one learned pair of numbers, but tuning one multiplier per feature by gradient descent worked far better. On 300 held-out CounterFact prompts it brought the true answer to rank 1 on 271, against 161 for a sweep given the same range and 94 for the original settings, with lower side effects and 5 to 6.5 times less compute. An additive edit then pushed random, unrelated words to rank 1 on 20 of 20 prompts, so reaching rank 1 says little about what a model knows. A side study applied the same tools to a new question: can an edit be carried from GPT-2 Small to GPT-2 Medium? A linear translator keeps the receiving model working. An exported edit lifts the target's median rank from 142 to 14 but reaches rank 1 on only 8% of records. What crosses is a push toward a word, not a fact. Mixing half of Medium's state into Small raises Small's accuracy from 8.5% to 9.5%. The side study's pass bars were fixed before the runs, and its limits (one model family, one dataset, no comparison with ROME or in-context editing) are stated.

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Chapter 1

Introduction

1.1 Background and Motivation

Language models often finish a factual prompt with a generic word even though the right answer is only a few places down the ranking. Given "The location of Massachusetts Institute of Technology is in", GPT-2 Small [1] puts " the" first and " Cambridge" eighth. We call this **suppressed factual recall**. The name only describes the output ranking; it does not say what the model "knows". The usual fixes change the model itself. Fine-tuning and closed-form weight edits such as ROME [2] and MEMIT [3] are permanent, and when they are applied again and again they first wear away earlier edits and then break the model [4]. Retrieval and in-context editing [32, 29] leave the weights alone, but they hand the fact to the model as extra text.

Mechanistic interpretability gives a third option. A sparse autoencoder (SAE) [6, 7] rewrites a layer's activations in a much wider basis where only a few directions are active at a time. These directions, called *features*, often line up with single concepts. A feature is a direction in the residual stream, so we can turn it down (*mute*) or up (*boost*) during one forward pass and throw the change away afterwards. If that works, a wrong answer can be repaired **transiently**, with no weight touched, and the repair can be read as a short list of named features. This project asks whether it works, how to build it, and what it actually does.

1.2 How the Project Grew

The project, **FeatureScalpel** (first called Causal Feature Steering), began in July 2026 as a small prototype and grew through trial and error. We logged every step in a numbered research journal, which had 37 entries by October 2026, and this report follows that journal in order. Figure 1.1 shows the whole path.

We want to be clear about how the pieces relate. The **main line** (stages 1 to 11) is the project itself: a transient SAE edit of GPT-2 Small, the question of how to choose that edit (a



Figure 1.1: The path of the project in twelve stages (J = research journal entries). Stages 1 to 11 are the main line (Chapters 3 and 4). Stage 12 is an applied side study (Chapter 5).

hand-built search, a trained network or gradient descent), and what the edit does apart from the one number it is tuned for. The **cross-model transfer study** (stage 12) is not a later version of the system. It is an *applied side study*. Once the edit, the controls and the benchmark existed, we used them on a new problem to see whether the interpretability and intervention work was useful beyond the single model it was built for.

1.3 Problem Statement and Research Questions

Suppose GPT-2 Small ranks the true answer to a prompt below first. Can a transient edit of SAE features during one forward pass bring the answer to the top, how should that edit be chosen, and what does reaching the top tell us? Five research questions follow from this.

1. **RQ1 (what works).** Which way of choosing features and strengths brings a suppressed answer to rank 1, and which ideas fail?
2. **RQ2 (limits).** What holds such an edit back: the depth of the intervention, the safety rule, or the supply of features it can use?
3. **RQ3 (choosing the edit).** Can a program choose the edit better than a hand-built search, either a trained network or gradient descent on the edit itself?
4. **RQ4 (meaning of success).** Does rank 1 show that a fact was recovered? What else does

the edit change: reworded prompts, neighbouring facts, unrelated prompts, generated text?

5. **RQ5 (application).** Can the change an edit makes in one model be carried into another model at inference time without changing weights, and what exactly is carried?

1.4 Objectives, Scope and Contributions

We set out to build a transient edit that changes only the intended feature contribution, to record with numbers what each design decision bought and which ideas failed, to build a benchmark with held-out prompts and exact paired statistics, to measure side effects against controls, and to apply the tools to the transfer question with pass bars fixed before each run. We use GPT-2 Small with the SAE release `gpt2-small-res-jb` (24,576 features per layer), layer 8 in all later studies, single-token facts from ROME's `known_1000` and CounterFact [2], and next-token measurements. The side study adds GPT-2 Medium, which has no SAE. Other models, comparison with ROME, MEMIT or in-context editing, and a full evaluation of generated text are outside the scope.

The main contributions are the following. (1) A record of the trial and error from single-feature ablation to gradient descent (Chapter 4). (2) A transient edit whose design choices are backed by measurements: delta patching, causal ranking, a strict per-feature safety filter, pool refill, and candidates from all prompt positions (10 of 13 prompts reach rank 1, against 4 with the last token only). A batched implementation cut model calls per prompt and layer from 186 to 6 with identical results. (3) An equal-budget comparison. On 300 held-out CounterFact prompts gradient descent reached rank 1 on 271, against 161 for a sweep given the same range and 94 for the original settings, with lower side effects on the same prompt and 5 to 6.5 times less compute. (4) A control showing that rank 1 is a weak test: an additive edit pushes random words to rank 1 on 20 of 20 prompts. This led us to withdraw an early "knowledge absent" classification. (5) An applied side study with fixed pass bars: a linear translator between GPT-2 Small and Medium, Export and Import tests with controls, a documented search that locates the limit, and a first positive Import result.

1.5 Organisation of the Report

Chapter 2 reviews related work, including where gradient-based optimisation of edits comes from. Chapter 3 gives the mathematics of the final system, Chapter 4 tells how we got there, and Chapter 5 covers the cross-model side study. Chapter 6 describes the application and the engineering, and Chapter 7 discusses the findings, the limits and future work. The appendices map journal entries to report sections, list the side study's pass bars and give the commands that reproduce the results.

Chapter 2

Literature Review

This chapter covers the work the project builds on, shows where our use of gradient descent sits in that literature, and states the gap we address. We read most sources through their abstracts and summaries rather than in full, and the numbers quoted are the sources' own.

2.1 Knowledge Editing

ROME [2] finds where a fact is stored in the middle MLP layers of GPT-style models and rewrites one weight matrix in closed form. MEMIT [3] extends this to thousands of edits. Both change the weights for good. Gupta *et al.* [4] showed that editing one fact after another slowly erases the earlier edits and then suddenly collapses the model, and Hase *et al.* [28] found that where a fact is stored does not necessarily tell you where it is best edited. ROME also gave us **CounterFact**, which pairs each prompt with a true and a counterfactual answer and adds reworded (paraphrase) and neighbouring (neighbourhood) prompts for testing generalisation and specificity. It also gave us the `known_1000` set of facts that its model already predicts, and like ROME we filter our facts in this way. Some methods leave the weights alone and add the fact as text instead. In-context editing [29] puts the corrected fact and a few demonstrations in the prompt, and retrieval-augmented generation [32] conditions the model on retrieved documents. Both change the input and not the model's own computation.

2.2 Activation Steering and Sparse Autoencoders

Activation steering adds a direction to the residual stream at inference time. Activation Addition [9] takes the difference between the activations of two prompts, contrastive activation addition [10] averages many such differences, and representation engineering [11] reads and controls directions that describe a whole population of examples. Superposition [5] means single neurons rarely match single concepts, so these directions tend to carry unrelated content with them. Sparse autoencoders [6, 7, 8] learn a wide basis in which only a few features are

active at once, and those features are often interpretable. Open SAEs exist for GPT-2 Small [15] and Gemma [17], with browsable feature descriptions [18].

Several studies edit SAE features directly. Chalnev *et al.* [26] aim steering vectors at particular features to limit side effects. Zhao *et al.* [27] edit a few features to decide whether a model relies on its memory or on the context. Farrell *et al.* [25] unlearn knowledge by ablating features, and Marks *et al.* [31] use features as the nodes of editable circuits. Arad *et al.* [21] show that picking features by their effect on the output improves steering two to three times, which is the idea behind our causal ranking. The evidence is mixed, though. AxBench [12] reports that simple baselines can beat SAEs at steering, and a Google DeepMind update [22] reports negative SAE results on downstream tasks. Chaudhary and Geiger [23] find that SAEs struggle to reach a neuron baseline on factual knowledge in GPT-2 Small, and Cui *et al.* [24] show that behaviour suppressed by clamping SAE features can come back through the part of the activation the SAE does not explain. None of these tests exactly our setting. Together they are the reason we measure side effects and use controls, and why we assume nothing about what a feature edit does.

2.3 Causal Localisation and Attribution

Activation patching measures how much an internal component matters for the output [19]. Trying every component is expensive, so gradient-based shortcuts exist: attribution patching and AtP* [16] estimate component effects from a single backward pass. We rank features by actually removing them, in batches (Section 3.3), and keep the gradient-based version in mind for larger fact sets. The logit lens [34] reads an intermediate state through the output projection, and we use it in our diagnostics.

2.4 Gradient-Based Optimisation of Edits and Steering Vectors

We came to gradient descent through our own data (Section 4.10). It is an old tool in this area, though, and the work below shows where our use of it fits.

- **ROME itself.** Before ROME changes any weight, it finds the vector to be written by gradient descent on the probability of the new answer, with a term that keeps the subject's meaning intact [2]. Our loss has the same two ingredients, reaching the target and limiting the change elsewhere. The difference is that we tune a few hundred SAE multipliers for one prompt and change no weight.
- **Soft prompts and prefix tuning.** Lester *et al.* [42] and Li and Liang [43] freeze the model and train continuous vectors by gradient descent, which shows that a frozen

model can be steered through a small optimised input.

- **Optimised steering vectors.** Subramani *et al.* [44] use gradient descent to find vectors that, added to GPT-2's hidden states, make it produce a chosen sentence almost perfectly. Dunefsky and Cohan [45] optimise a steering vector from a single example and study how far it generalises, including to harmful behaviour.
- **Adversarial examples.** Szegedy *et al.* [46] and Goodfellow *et al.* [47] showed that a small perturbation found by gradient descent can push a network to any output, and Zhang *et al.* [48] showed that networks can fit random labels. This is the right comparison for our additive edit (Section 4.11): once an optimiser has enough freedom, reaching the chosen output proves little.

As far as our limited search could tell, none of this work tunes one multiplier per *SAE feature* for a single prompt with a side-effect term, or compares that with a hand-built feature sweep on held-out prompts.

2.5 Comparing and Stitching Representations

Model stitching joins the lower layers of one network to the upper layers of another through a small learned layer. If the stitched network still works, the two representations are compatible [37, 38]. Similarity measures such as CKA [39] compare activations without stitching. For language models, Chen *et al.* [40] show that a simple affine map between the residual streams of two models moves features across at low cost. They carry SAEs from a smaller to a larger model, which cuts the larger SAE's training cost by about half, and they recover probes and steering vectors. Our translator (Chapter 5) is a ridge regression [41] of the same family, and we do not claim it as new.

2.6 Research Gap

Knowledge editing changes weights or the prompt, SAE steering works inside one model, and stitching work moves vectors learned in one model into another. As far as a limited search could tell, we did not find the following together: a joint mute-and-boost edit of causally ranked SAE features for a single fact, with a per-feature safety filter and pool refill; a comparison on held-out CounterFact prompts, with one yardstick, between a hand-built sweep and per-feature gradient descent; a control showing that an additive edit pushes unrelated words to rank 1; and a test of whether the change an edit makes in one model can be carried into another, with pass bars fixed beforehand. This report supplies these, negative results included. It does not claim that the SAE route beats prompting, probes, difference-of-means steering, ROME or in-context editing, because that comparison has not been made.

Chapter 3

The Edit and How It Works

This chapter gives the mathematics of the final system as it is implemented in `src/hooks.py`, `src/editing.py`, `src/hybrid_runner.py` and `src/gradient_editing.py`. Chapter 4 then shows how each piece came about.

3.1 Setting and Notation

GPT-2 Small reads a prompt and, at the last position, gives a probability for every possible next token. The **residual stream** is the running sum that every attention and MLP block adds its output to. We write $x_p \in \mathbb{R}^{768}$ for its value at position p at the hook point `blocks.L.hook_resid_pre`, with layer $L = 8$ unless stated otherwise. For a prompt whose true answer is t, the **rank** of t is its place in the next-token ordering (1 is first), and the **blocker** b is the token currently ranked first. An edit **succeeds** when t becomes first. That is a low bar (Section 4.11), so we always read it together with side-effect measures and controls.

3.2 SAE Features and Delta Patching

For the activation x_p the SAE gives sparse feature activations and a reconstruction,

$$f_p = \mathrm{ReLU}\big(W_{\mathrm{enc}}(x_p - b_{\mathrm{dec}}) + b_{\mathrm{enc}}\big) \in \mathbb{R}^{24576}, \qquad \hat{x}_p = W_{\mathrm{dec}}^{\top} f_p + b_{\mathrm{dec}}, \tag{3.1}$$

where row $W_{\mathrm{dec}}[k]$ is the direction of feature k in the residual stream. If we simply replaced x_p by $\hat{x}_p$ we would inject the SAE's reconstruction error (12 to 26% of the activation norm) and disturb the model even with no edit. So we add only the change in the decoded output. If feature k is scaled by s_k, with $s_k = 1$ for features left alone, then

$$\Delta_p = \sum_k (s_k - 1)\, f_{p,k}\, W_{\mathrm{dec}}[k], \qquad x'_p = x_p + \Delta_p, \tag{3.2}$$

at every prompt position p. Muting with strength $\mu \in [0,1]$ means $s_k = 1-\mu$, so $\mu = 1$ removes the feature, and boosting with strength $\beta \geq 0$ means $s_k = 1+\beta$. When every $s_k = 1$ the edit is exactly zero.

3.3 Choosing Candidates: Causal Ranking

Ranking all 24,576 features by removing each one would take as many forward passes. We do it in two stages instead. **Stage 1** screens. One clean pass caches the residual stream, the SAE encodes it, and we keep the top N features by activation, either at the last token or over *all* prompt positions except the start token, scoring each feature by its maximum activation. **Stage 2** ranks by cause. Each candidate is removed on its own and we record how a token's probability changes,

$$\Delta p_k(v) = P_{\text{ablate}(k)}(v) - P_{\text{clean}}(v). \tag{3.3}$$

For the blocker ($v = b$) the most negative values are its strongest drivers, which form the *mute list*. For the target ($v = t$) the most negative values are its strongest helpers, which form the *boost list*. The N removals are stacked into the batch dimension of a single model call, so ranking costs two batched calls instead of $2N+1$ separate ones.

3.4 The Hybrid Mute and Boost Sweep

The **safety filter** tests every candidate alone at its strength and keeps it only if the target does not get worse:

$$\text{safe}(k) \iff r_k(t) \leq r_0(t) \ \wedge \ p_k(t) - p_0(t) \geq -10^{-6}, \tag{3.4}$$

where r and p are the target's rank and probability with feature k applied alone, and r_0, p_0 are the baseline values. A feature that is safe on both sides is kept on one side only. The sweep then applies the first k mutes and the first k boosts together in one forward pass, for $k = 1, 2, \ldots$, and stops when the target is first. It reports the best step and not the last one, because adding features that are each safe does not always help. When both pools are used up, **pool refill** keeps every feature applied so far, makes one pass to get the *steered baseline* and its new blocker, and ranks and filters the remaining features again, including ones rejected earlier.

3.5 The Gradient-Descent Editor

The sweep decides which features to use and how many, with one strength per side. The gradient-descent editor instead gives each of $K = 200$ candidate features its own multiplier and tunes all of them for the prompt at hand. With free numbers z_k, set at the start so that no

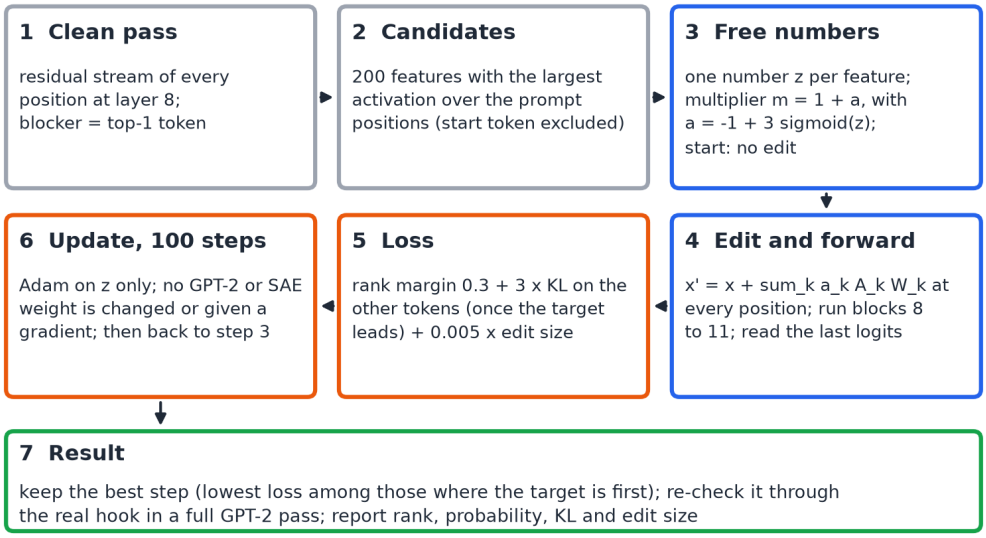


Figure 3.1: The gradient-descent editor. Only the free numbers z_k are updated.

feature changes, the multiplier is $m_k = 1 + a_k$ with

$$a_k = -1 + 3\,\sigma(z_k), \qquad m_k \in (0,3), \tag{3.5}$$

where σ is the logistic function. With $A_{p,k}$ the clean activation of feature k at position p, the edit is Equation 3.2,

$$\Delta_p = \sum_{k=1}^{K} a_k A_{p,k} W_{\text{dec}}[k]. \tag{3.6}$$

We run blocks 8 to 11 from the edited state and read the logits ℓ at the last position. The loss is

$$\mathscr{L} = \max\bigl(0,\ \gamma - (\ell_t - \max_{v \neq t} \ell_v)\bigr) + \lambda_{\text{KL}}\,\mathbf{1}[\text{target leads}]\,\text{KL} + \lambda_{\text{size}} \sum_k |a_k|\, w_k, \tag{3.7}$$

with margin $\gamma = 0.3$, $\lambda_{\text{KL}} = 3$, $\lambda_{\text{size}} = 0.005$, and w_k the feature's share of the total maximum activation. The first term asks the target to beat every other token by γ logits. The second penalises changes to the rest of the distribution,

$$\text{KL} = \sum_{v \notin \{t,b\}} q_v \log \frac{q_v}{\tilde{p}_v}, \qquad q = \text{softmax}\bigl(\ell^{\text{clean}}_{\setminus\{t,b\}}\bigr),\ \ \tilde{p} = \text{softmax}\bigl(\ell_{\setminus\{t,b\}}\bigr), \tag{3.8}$$

taken over all tokens except the target and the blocker, so that the intended change is not penalised. It is charged only once the target leads, which matches the sweep's stopping rule. The third term prefers small edits. Only the z_k are updated, by Adam [33] (learning rate 0.1, 100 steps), and GPT-2's weights and the SAE are frozen and receive no gradient. We keep the best iterate and re-check it through the same hook as the sweep in a full forward pass, so the reported rank does not depend on the shortcut used during tuning. Figure 3.1 shows the flow.

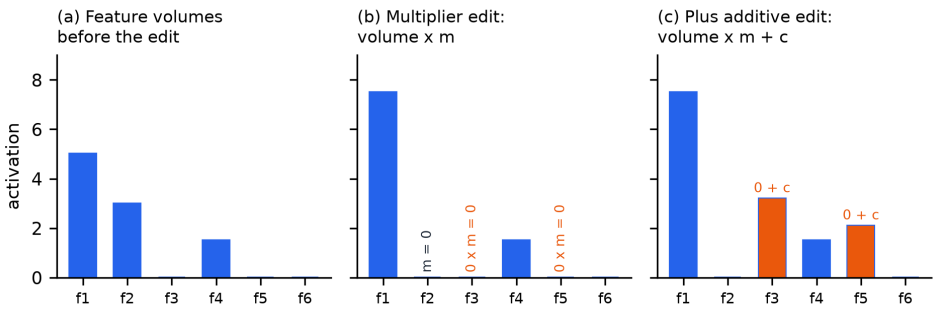


Figure 3.2: Why an additive edit was tried. A multiplier leaves a silent feature (f3, f5) at zero, while an additive amount c switches it on.

3.6 The Additive Edit

A multiplier cannot switch on a silent feature, because $0 \times m = 0$ (Figure 3.2). The additive extension adds an amount $c_k \geq 0$ of the decoder direction of selected silent features, at the last position only:

$$\text{new volume} = \text{old volume} \times m + c, \qquad \Delta_{\text{last}} \mathrel{+}= \sum_{k \in \mathscr{S}} c_k W_{\text{dec}}[k]. \tag{3.9}$$

The set $\mathscr{S}$ holds the 200 features that are exactly silent at the last position and have the largest positive first-order score $W_{\text{dec}}[k] \cdot g$, where $g = \partial(\ell_t - \max_{v \neq t} \ell_v)/\partial x_{\text{last}}$ is the gradient of the target's margin with respect to the residual. Each amount is $c_k = \text{cap} \cdot (\sigma(s_k) - \sigma(s_0))$ for $s_k \geq s_0$ and 0 otherwise. Every amount starts at exactly zero, cap is the loudest feature on the real prompt positions, and an extra loss $0.005 \sum_k c_k/\text{cap}$ discourages large additions. The score is a first-order estimate of how the margin changes. That is why this edit writes the *output direction of the answer* and not a representation of a fact (Section 4.11).

3.7 Benchmark and Metrics

The benchmark (16,360 CounterFact prompts, a subject-safe split, 300 test prompts) is described in Section 4.8. We report the *rank* and *probability* of the target; the *same-prompt KL* (Equation 3.8); the *edit size*, which is the norm of the change divided by the norm of the residual at the last position; and the effect of the same edit on 20 unrelated prompts, on CounterFact's reworded and neighbouring prompts, and on generated text. Methods are compared on the same prompts with the exact McNemar test [35] on reaching rank 1 and the Wilcoxon test [36] on KL. We made our choices on validation prompts and only read the test prompts.

Chapter 4

Evolution of the System: a Record of Trial and Error

This chapter tells how we arrived at the system described in Chapter 3. For each stage we say what we tried, what happened and why we moved on, following the order of the research journal (Appendix A maps journal entries to sections). The early experiments used one prompt or 22 facts. They explain why the design looks the way it does, but they are not performance estimates.

4.1 The Starting Point

The first prototype (July 2026, journal 1) loaded GPT-2 Small and the layer-8 SAE and added a *causal selector*. It took the features active on the last token of the prompt, removed each one in turn, and ranked them by how much the target's probability changed. Three things we learned there shaped the rest of the work. The most active feature is not the one that changes the answer, so we rank by causal effect and not by activation. Removing a feature completely damaged other predictions, whereas a partial change (strength 0.2 to 0.3) did not on the one example we tried. And an *eligibility rule* applied to 45 fact pairs from ROME's `known_1000` gave 15 already correct, 22 *suppressed* (target not first but probable) and 8 improbable. The roadmap we wrote then already listed a learned predictor of which features to edit, which we reached in Section 4.9, and a comparison with ROME, which is still open.

4.2 Single-Feature Ablation

Our first search ablated, one after another, the features that *help* the target (strength 0.3, up to five rounds) on the 22 suppressed facts (journal 2). It corrected **1 of 22**. The reason is simple in hindsight: muting what helps the target weakens the target. The second version muted the drivers of the current *competitor*, the token ranked first. It corrected **3 of 22**, each in one round

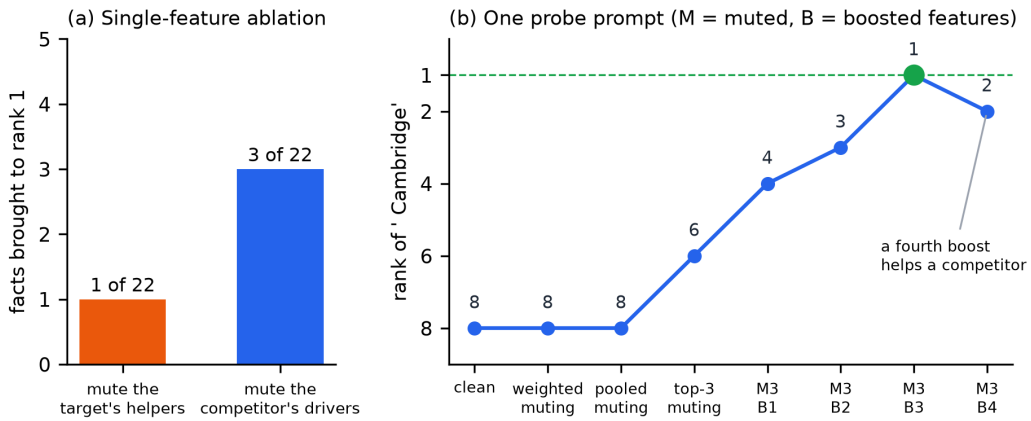


Figure 4.1: The early experiments. (a) 22 suppressed facts: muting the target's own helpers corrected 1, muting the competitor's drivers corrected 3. (b) The probe prompt: muting alone stayed at rank 8, mute and boost reached rank 1, and a fourth boost undid it.

and with no change to 15 control prompts (journal 3). All three started at rank 2, though, and the method failed whenever the leader was a generic word such as " the" (Figure 4.1a). One feature was not enough, and we wanted to know why.

4.3 Several Competitors, Spread-Out Support

For a probe prompt ("The location of Massachusetts Institute of Technology is in", target " Cambridge", rank 8, with seven words above it) we muted all the competitors in proportion to their probability (journals 4 to 8). The seven competitors collapsed onto just two features: 313 for " the", " a" and " an", and 21169 for " question", " jeopardy", " danger" and " doubt". The target stayed at rank 8. Raising the strength on feature 313 by 73% moved the target's probability only from 1.29% to 1.39%, so muting too weakly was not the problem. When we looked at each competitor's top ten features, the semantic competitors turned out to be spread over many of them, with the second feature 92 to 95% as influential as the first. Muting the top three features per competitor reached rank 6, and five diluted the effect back to rank 7.

4.4 Mute and Boost Together

Adding the opposite move, boosting the features that support the target, changed things at once. Three mutes and three boosts brought the probe prompt to **rank 1** (journal 5; Figure 4.1b). A fourth boost pushed it back to rank 2. This was not a bug, since an isolated rerun gave the same 6.7357%. The fourth feature supports " the" more than " Cambridge", and because probabilities pass through a softmax, more editing is not always better. The lesson was to report the *best* step and not the last. The hybrid edit with a safety filter (rank 2 on this prompt, safer but weaker) became our working algorithm.

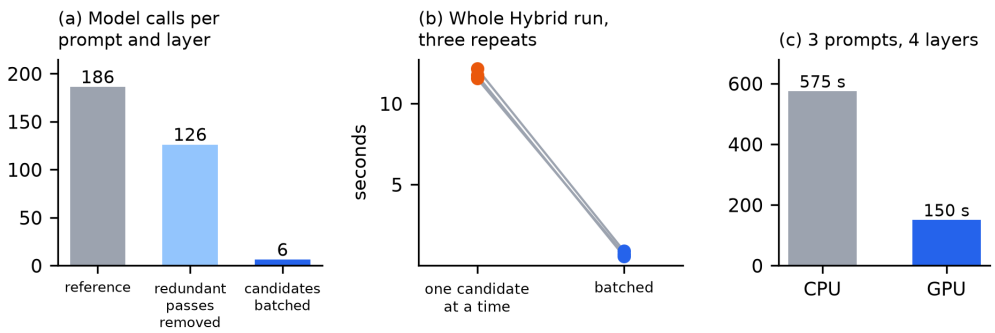


Figure 4.2: Making the computation fast. (a) Model calls per prompt and layer. (b) Whole Hybrid run with the same result, three repeats. (c) CPU against GPU on 3 prompts and 4 layers.

4.5 Where to Edit, and Making It Fast

A layer sweep on 21 prompts (12 of them eligible) showed a plateau at layers 6 to 9, with layer 8 highest. Its mean rank gain was 7.33, against 7.08, 6.50 and 6.33 at layers 7, 6 and 9, and at most 4.75 at the other layers (journals 10 and 11). We kept layer 8 as a reasonable default, not as a proven optimum.

The first real engineering step was a cost audit. A single prompt at one layer needed **186 model calls**. Both safety filters were recomputing the clean baseline inside every candidate loop, and passing it in removed 60 calls with identical results. Stacking the candidates into the batch dimension of one call then brought the number down to **6**, a 21-fold reduction. The ordered feature lists, the probabilities (to 10^{-9}) and all 60 safety decisions matched the sequential code, which we kept as the reference. A CUDA GPU gave another factor of 3.8, and in the application the whole Hybrid run fell from about 12 s to under 1 s (Figure 4.2).

4.6 Making the Sweep Reliable

Once the sweep was fast, its weaknesses became visible (journals 16 to 19). It reported the last step instead of the best one, which we fixed with best-so-far tracking. Applying mute and boost as two chained hooks gave rank 66 where the combined safety check gave 59, so we switched to a single combined hook. And it stopped early, because its pool held only features that had passed the filter against the *clean* model. **Pool refill** fixed that by re-ranking against the edited model, and it exposed a hard limit. For “The leading cause of death in men is” only 67 features were active at the last token, and scaling cannot activate a feature that is off. The run used all 67 and stopped at rank 51 (from 152).

The fix was to draw candidates from **all prompt positions**, since the meaning of “Sophia” sits on the word *Sophia* and not on the last word of the prompt. On 13 comparable prompts the all-positions version reached rank 1 on **10**, against 4 for the last token. It won on 9 prompts

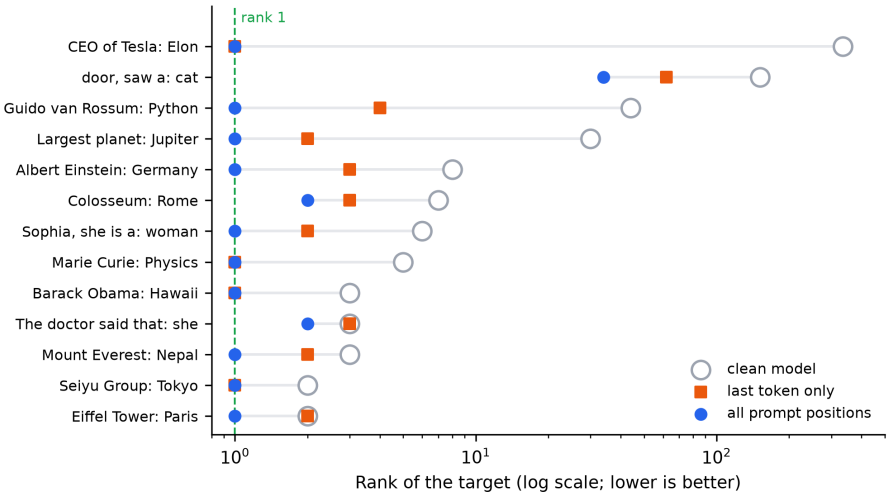


Figure 4.3: Final rank of the target on the 13 comparable prompts when candidates come from the last token only or from all prompt positions (log scale; rank 1 is the dashed line).

and lost on none (sign test $p \approx 0.004$), and in all 9 prompts where the last-token version failed, it had used every feature it could see (Figure 4.3; journal 20). We then asked whether the strict safety rule was too strict. In a 45-prompt pilot, removing the filter lost 10 of 29 successes and gained none, a 5% tolerance lost 2, and graded strengths changed nothing. On 131 last-token prompts, relaxed rules added at most one prompt. Only 2.5 to 2.9% of features were rejected on both sides, so the supply of features, and not the filter, was the real limit, and the strict filter stayed.

4.7 Is the Layer, or the SAE, the Limit?

We tested three worries on the prompts that had failed in every mode (journal 22). Later layers might rebuild what the edit removes. The SAE might be more machinery than a factual edit needs. And the SAE's reconstruction error, which we never edit, might hold the missing information. The pushed direction was still 0.79 to 1.29 of its injected size at the output, and re-adding whatever fraction had been lost gave the same rank in 21 of 21 layer and prompt cells, so **we saw no repair**. A free residual edit of the same size reached rank 1 in all 21 cells, against 1 for the SAE edit. We chose these prompts because the SAE edit had failed on them, so this is a yardstick and not a success rate. A literature check the same week (journal 25) found mixed evidence on SAE steering and led us to compare every later result with controls.

4.8 Scaling Up: the CounterFact Benchmark

Our hand-assembled sets (22, 45 and 131 prompts) share templates and are too small for held-out tests, so we built a benchmark from CounterFact (journal 24). Of its 21,919 records, 16,360

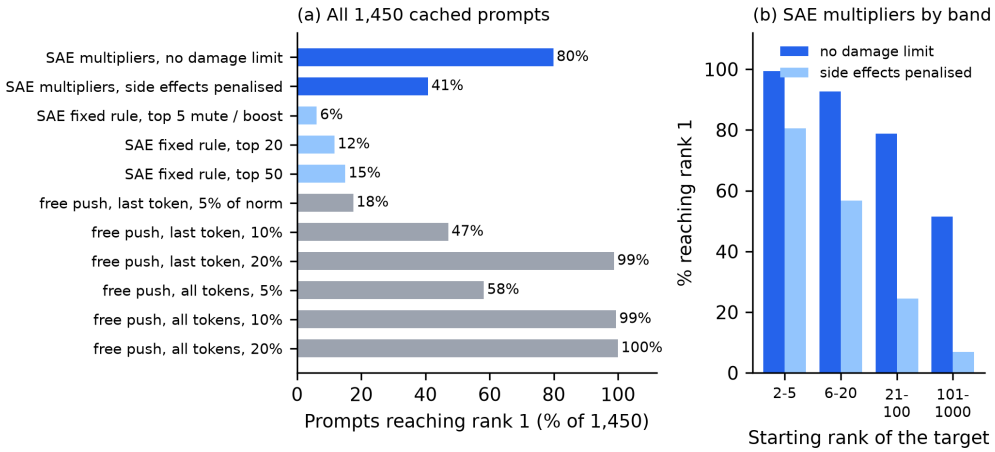


Figure 4.4: Headroom on the 1,450 cached prompts (one-shot edits on saved states, a stand-in for the real sweep): (a) share reaching rank 1 by method; (b) free multipliers by starting-rank band.

have a single-token true answer that GPT-2 Small ranks between 2 and 1000. We split them so that a subject appears in only one set and five relations are held out, which gave 1,000 training, 150 validation and 300 test prompts. For 1,450 of these prompts we cached the layer-8 states, the candidate features, their removal effects and the safety filter's verdict at ten strengths, once and for all. A **headroom** measurement then asked how much room any choice of multipliers has. Free per-feature multipliers could bring the target to rank 1 on **80%** of the 1,450 prompts if side effects are ignored, and on 41% with a side-effect penalty, against 6 to 15% for crude fixed rules (Figure 4.4).

4.9 Learning the Strengths

The sweep uses one mute strength (0.6) and one boost strength (0.5) for every prompt, even though the strengths change the outcome. We trained a small network to choose them (journals 23, 26 and 27). *PromptNet* reads 12 summary numbers about the prompt, such as the target's rank and probability, and outputs a mute and a boost strength. *FeatureNet* outputs one multiplier per candidate. Training back-propagated through blocks 8 to 11 from the cached states. We compared them with a *learned fixed pair* (two numbers shared by all prompts) and a *per-prompt oracle* tuned by gradient descent at test time. On the 300 test prompts, under a stand-in measure, fixed 0.6/0.5 reached rank 1 on 75 (25.0%), the learned pair (mute 0.98, boost 1.81) on 116 (38.7%), PromptNet on 119 and the oracle on 123 (Figure 4.5). The gain came from a better *global* pair. Choosing per prompt added only 3 prompts (exact McNemar $p = 0.25$), side effects roughly doubled, and FeatureNet (31 to 32%) stayed below the pair. We never ran the real sweep with the learned strengths. The network turned out not to be the useful part. Its *teacher* was: free multipliers tuned on each prompt reached 73% in the same stand-in, nearly twice what the network managed.

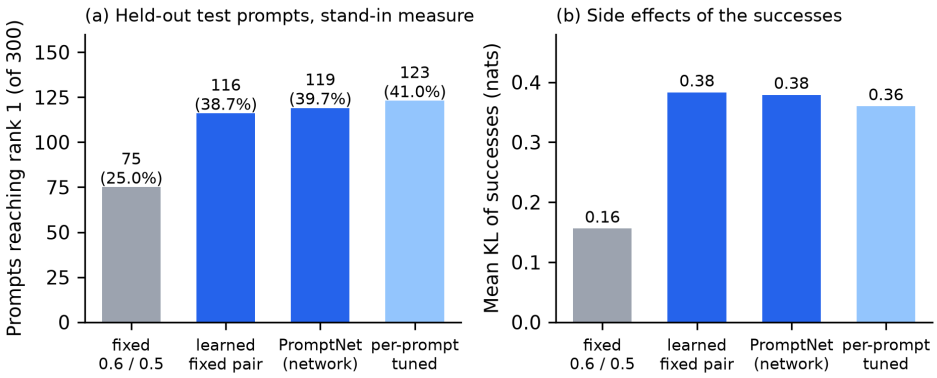


Figure 4.5: Learned strengths on the 300 test prompts (stand-in measure): (a) prompts reaching rank 1; (b) mean KL of the successes.

4.10 Gradient Descent Enters

Gradient descent was in the project for a week before it became the method. It first appeared as a measuring stick, as the bound for an edit made directly in the residual stream (journal 22, Adam on a free vector). It came back in the headroom measurement (journal 24) and in the oracle (journal 27). The oracle's 73%, against 31 to 32% for the network, suggested using the optimiser itself as the editor (journal 28, October 3). It gives one multiplier per candidate, tuned for the prompt by 100 Adam steps, in about 6 to 8 s per prompt. Section 2.4 traces the idea in the literature.

The first hand trials favoured gradient descent but were unfair, because it could scale features from 0 to 3 times and the sweep only from 0.4 to 1.8. So the controlled comparison (journal 29) gave the sweep the same range (mute 1.0, boost 2.0) and measured all arms with one yardstick through the real hook. On the 300 test prompts gradient descent brought the true answer to rank 1 on **271** (90%), the wide sweep on 161 and the original sweep on 94 (Figure 4.6). It solved 110 prompts the wide sweep missed and lost none. Its same-prompt KL was lower on 160 of the 161 prompts that both solved, and it needed 7.7 s per prompt against 38.5 and 49.7 s. Two cautions apply. Part of the KL advantage is built in, since its loss contains a KL term and the sweep has none. And the median target probability at rank 1 is only 7.7%.

4.11 Forcing Every Target to Rank 1: the Additive Edit

The 29 prompts that gradient descent failed on are mostly deep targets whose helpful features may be silent, and a multiplier cannot switch on a silent feature. The additive edit of Section 3.6 can, and its first hand trials looked wonderful. All six hard prompts we tried reached rank 1, including absurd targets such as a random name after "Hello, my name is". We ran a control that paired 20 validation prompts with random, unrelated words. Multipliers reached rank 1 on 1 of 20, but the additive edit reached it on **20 of 20**, using an edit 126 to 192% the

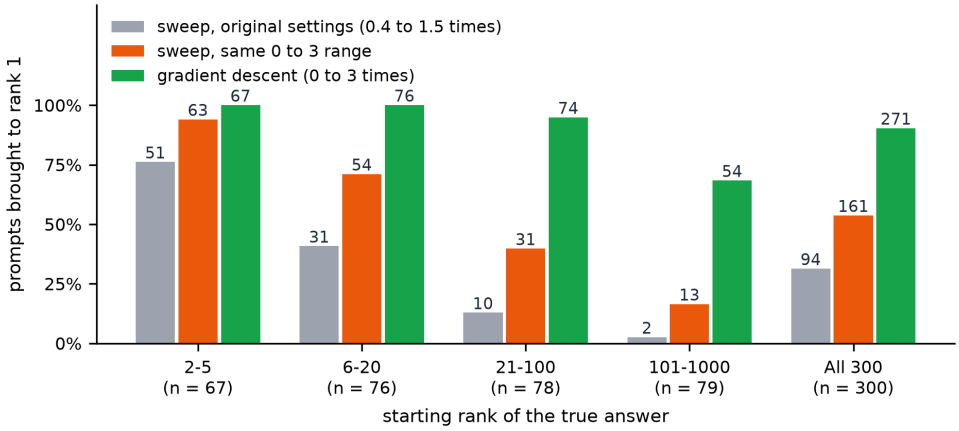


Figure 4.6: Prompts brought to rank 1 by starting rank of the true answer, 300 held-out prompts: the original sweep, the sweep with the same 0 to 3 range, and gradient descent (green).

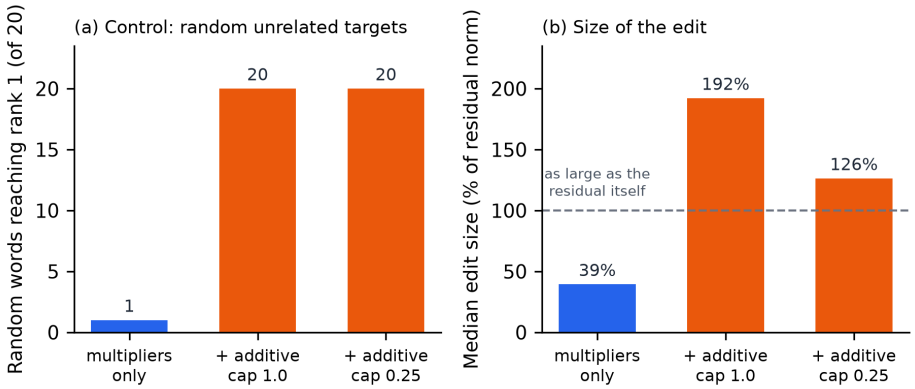


Figure 4.7: Control with random unrelated targets: the additive edit reaches rank 1 on 20 of 20 prompts at a size above the residual stream's, and multipliers on 1 of 20.

size of the residual stream itself (Figure 4.7). This is the analogue of fitting random labels [48] or of an adversarial perturbation [46, 47]: with enough freedom, an optimiser reaches any output you ask for. The edit also carried over to reworded prompts about as well for a random word as for the true answer (median rank 4 against 4.5, and rank 1 on 25% against 16%), and it changed the top word on 82% of unrelated prompts. The edit *steers toward a word* and does not repair a fact.

Three things followed. Rank 1 is a weak test, so every later result needs controls. Being able to steer a target to the top tells us nothing about knowledge in either direction, so we withdrew the early label "knowledge absent" as a statement about the model. And we switched the additive edit off by default and added a warning. It answered the question "can we force GPT-2 to say what we want?" with "almost anything", which is exactly why it is not a fix.

Table 4.1: What the trial-and-error record kept and what it dropped.

Kept	**Dropped, and why**
Delta patching: reconstruction error never injected	Muting the target's helpers: weakens the target
Causal ranking by ablation, in batches	Weighted, summed or top-K muting alone: support is spread
Mute and boost with a strict per-feature filter	Relaxed or graded filters: at most one prompt gained
Best-step reporting, pool refill, candidates from all positions	Last-token candidates: features run out
Gradient-descent editor (271 of 300)	Learned strength network: no gain per prompt
Controls, held-out benchmark, fixed pass bars	Additive edit as a fix, and the "knowledge absent" label

4.12 What the Edit Does Beyond the Next Word

We then tested the multiplier edit on 40 test prompts (10 per band) for what it does besides reaching rank 1 (journal 30). For *reworded prompts*, we tuned the edit on one prompt and applied it unchanged. It lowered the true answer's median rank from 94 to 36 and reached rank 1 on 9% of the rewordings, while a random word went from 17,504 to 11,832 and reached it on none. So there is a modest fact-related part, but the edit is mostly local. On *unrelated prompts* the multiplier edit changed the top word of 6% of 20 prompts, and the additive edit 17 to 24%. For *generated text* (20 greedy tokens, edit kept on), fluency was unchanged (perplexity 4.4 to 4.8 against 4.7), variety dropped slightly (distinct 2-grams 0.85 to 0.89 against 0.93), loops appeared in 2 to 12% of continuations against none, and the target word was repeated about 0.6 to 0.7 extra times. For *instructions*, we added the prefix "Do not say the word X". The unedited model already put X first on 45% of prompts, because a named word primes copying. The multiplier edit raised this to 68% and the additive edit to 88 to 90%, so the edit ignores the instruction.

4.13 Where the Main Line Stands

Table 4.1 sums up the record. A transient SAE edit brings a suppressed answer to the top on most held-out prompts, cheaply and with small side effects on the same prompt. It does this by pushing toward a word, it carries little beyond the prompt it was tuned on, and it ignores instructions. Three limits remained. An edit is tied to one model that has an SAE, GPT-2 Small lacks answers that larger models hold, and rank 1 needs controls to mean anything. These led us to a question for the tools we had built: could the change an edit makes in one model be carried into another? Chapter 5 reports that study.

Chapter 5

Applied Study: Carrying an Edit to Another Model

This chapter describes a *side study*. It is not the next version of the system from Chapters 3 and 4. We took the tools that work had produced (the edit, the benchmark, the controls and the habit of fixing pass bars first) and applied them to a new question, to see whether the interpretability and intervention work was useful beyond one model.

5.1 The Question and the Design

Can the change an edit makes inside GPT-2 Small be translated and added to GPT-2 Medium at inference time with no weight change, and can Medium's state improve Small? We call the first direction **Export** (small to medium) and the second **Import** (medium to small). Figure 5.1 shows both. The two models do not share internal coordinates (width 768 against 1024), so we need a *translator*. And since Section 4.11 showed that "the answer reaches rank 1" is weak evidence, every test here has controls.

Models and data. "Small" is GPT-2 Small with its layer-8 SAE, and "medium" is GPT-2 Medium (24 layers, width 1024, no SAE). They share one tokenizer. We pair small's layer 8 with medium's layer 16. The benchmark is 200 CounterFact records from the hard set of Section 4.8, split into 50 development (dev) records and 150 test records. Dev records are used to choose settings (layer, dose, epoch), and test records are only read. The edit tuned in small reaches rank 1 on 116 of the 200 records. The **88 primary test records** are the test records where it does, because only there is there an edit to carry over. Before anything is done, medium's median rank for the target is 142.

The translator. We translate only the *change* an edit makes, so the bias of an affine map cancels and the translated change is $\Delta_S W$ (zero at the first position). We tried four ways of getting W. (1) A *ridge map* fitted on ordinary text, $W = \arg\min \|(H_S - \mu_S)W - (H_M - \mu_M)\|^2 + \lambda \|W\|^2$, over paired states of the two models on wikitext-2 (500,126 fit tokens) [41].

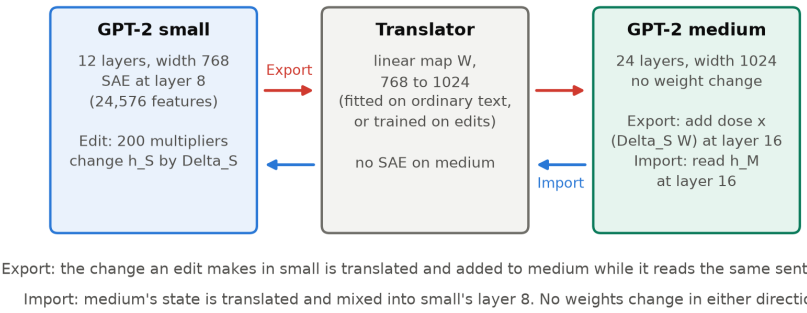


Figure 5.1: The two directions studied. A translator maps the residual stream of one model into the other's coordinates. No weight changes, and the SAE is used only where the edit is made.

(2) A *neural map*, which is the ridge map plus a one-hidden-layer correction. (3) A *linear map trained on edits*, which starts from the ridge map and is trained so that medium's answers move the way small's do. Its goal is $q = \mathrm{softmax}(\ell_M + \ell_S^{\mathrm{edit}} - \ell_S)$ and its loss is $\mathrm{KL}(q \,\|\, p_M(h_M + \alpha \Delta_S W))$ over 289 successful training edits. (4) *No fixed map*: we re-tune the 200 multipliers for each sentence against *medium's* output. This is a ceiling and not a transfer, since it looks at medium every time.

Export, Import and controls. Export adds the translated change while medium reads a sentence, $h'_M = h_M + \alpha \Delta_S W$ at layer 16 at every position except the first. The **dose** α is chosen on dev. From step D10 on we chose it by the best mean log-rank gain, because choosing by dev top-1 on 28 records proved fragile. Import mixes small's state with medium's translated state, $h'_S = (1-\gamma)h_S + \gamma T(h_M)$, where $\gamma = 1$ means full replacement. The receiving model is never shown the answer. The controls are a random vector of the same size, another record's edit, an edit toward a random word of matched starting rank, a *word push* along the target's output direction, and the fact stated in medium's prompt. The word push uses the answer, so it is a comparison and not a method. For Import the controls are a state from a wrong sentence, the average state and a random map. *Rewordings* and *neighbours* are the share of reworded prompts on which the new answer beats the true one, and the share of other-subject prompts on which the true answer still wins. **Every pass bar ("gate") was written into the project plan before the run that judged it, and none was moved.** A step designed after we saw a result is labelled post-hoc and cannot change a verdict (Appendix B).

5.2 The Channel Works

The ridge map fits well enough to keep the receiving model working. Per-dimension R^2 is 0.58 from small layer 8 to medium layer 16 and 0.66 from medium 16 to small 8. When we stitch one model's first layers to the other's last layers through the map, it recovers 91% (Export) and 98% (Import) of the receiver's own quality. The map copies *which word changed* well (cosine

Table 5.1: Export of the edit through the ridge map (88 primary test records; dose chosen on dev).

Arm	Top-1	Median rank	Rewordings	Neighbours
Medium unchanged	0%	142	36%	67%
Translated edit	**8%**	**14**	**52%**	**63%**
Random vector	0%	157	38%	66%
Another record's edit	0%	128	36%	65%
Word push (uses the answer)	83%	1	82%	23%
Fact stated in medium's prompt	–	–	97%	44%

0.63 to 0.76) but the summary at the last position only weakly (0.24 to 0.31). The translator therefore passes its pre-set bar for Import and misses it for Export.

As a positive control we swap one country for another in a template sentence. We translate the difference between the source model's states, add it to the receiver, and count a hit when the receiver's top answer becomes the swapped capital. The translated change works on **55%** of test cases for Export and **33%** for Import, against at most 0.4% for a random vector, a random map and another pair's difference.

5.3 Export: a Real but Weak Effect

For each of the 88 primary test records we take the edit tuned in small, pass it through the ridge map and add it to medium at layer 16 (Table 5.1). The effect is real, graded and specific. The target's median rank in medium falls from 142 to **14** and it becomes the top answer on **8%** of records. Reworded prompts favour the new answer on 52% against 36% unchanged, and neighbours lose only 4 points. A plain word push makes the target first on 83% of records, but only 23% of neighbours keep the true answer, and stating the fact in medium's prompt costs 23 points of neighbours. The pre-set bar (top-1 of at least 15%, and 10 points above the strongest control) was not met.

5.4 Searching for a Better Translator

Since the effect was real but weak, we tried seven changes, each with its pass rule written first (Figure 5.2; Appendix B). Injecting at layers 4, 8, 12, 16 and 20 gave 0, 5, 5, 8 and 1%, and splitting the dose over several layers did no better, so the layer is not the limit. A neural map fits ordinary text much better (R^2 up by 0.09 to 0.11) but exports no better (8%), so a better fit to text does not help. A **map trained on edits** (method 3) roughly doubled the result to 18% top-1 (median rank 14 down to 7, paired $p = 5 \times 10^{-9}$) and stayed specific to the record. More training edits, longer training and a direct rank-1 term in the loss all left it at about 20%.

Why it plateaus. With the rank-1 term, the map learned its 289 training targets (95%

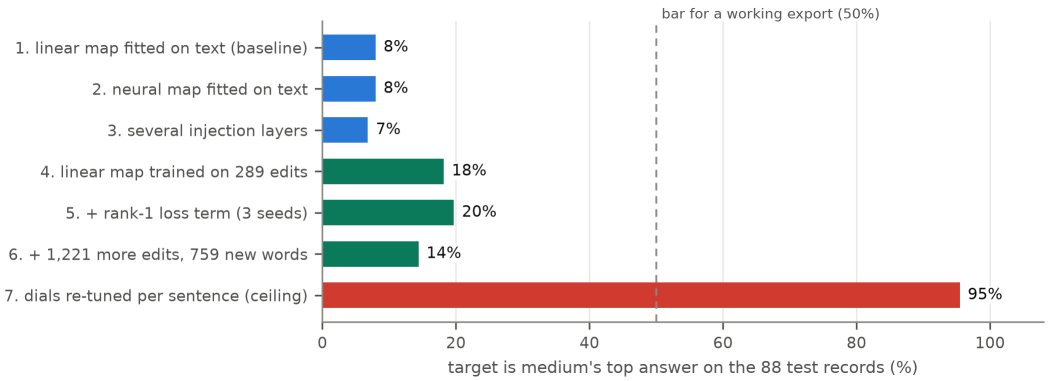


Figure 5.2: Top-1 of every export variant on the same 88 test records. Blue: map fitted on text. Green: map trained on edits. Red: per-sentence tuning against medium's output (a ceiling, not a transfer).

top-1 on them) but not unseen words (16%), which means it memorises. When we gave it 1,221 more edits toward 759 random target words, the memorisation went away but unseen records got *worse*, from 20% to 14%. The loss asks medium to copy small's shift and not to reach rank 1, so a better copy does not help. **Per-sentence tuning is a ceiling.** Re-tuning the multipliers against medium's output reaches the top answer on 95% of records (84% on all 150), with the same neighbour cost as the source edit inside small (8 points against 9). So the channel can carry a rank-1 edit, and what fails is the compatibility of an edit tuned for small with medium. The ceiling's solution points only partly in the direction of the original edit (cosine 0.51 against 0.02 by chance), so a map that merely transforms the given edit cannot supply what is missing.

5.5 What Is Carried: a Fact or a Push?

The question from Section 4.11 applies to Export too. The gain on reworded prompts has to be measured against each arm's *own* unchanged level, since a random word starts lower (25% against 36%). Measured this way, the per-sentence multipliers gain 24.4 points for the real target and 17.0 for a random word (difference +7.4, 95% interval 0.0 to +15.3, $p = 0.031$). The map trained on edits gains 19.9 and 18.2 ($p = 0.36$). Neither difference passes the $p < 0.01$ line we fixed beforehand. What carries over to reworded prompts is a push toward the targeted word, not something specific to the fact. In one more control we repeated the per-sentence tuning with 200 random directions instead of the 200 SAE directions. The SAE directions reached 95% and the random ones 81%. So the SAE makes the tuning more effective, but it is not what makes steering possible.

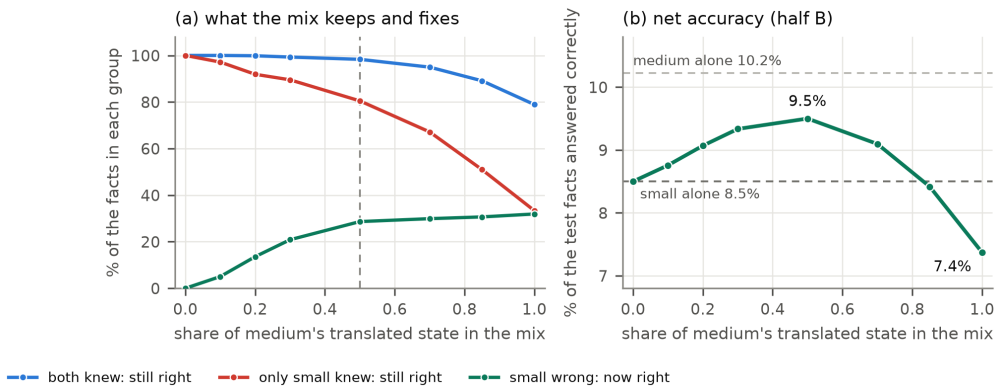


Figure 5.3: Import by mixing small's own state with medium's translated state. (a) What the mix keeps and fixes. (b) Net accuracy, which is best at a 50/50 mix and falls below small alone under full replacement.

5.6 Import: Medium's State into Small

We use the 21,789 single-token CounterFact facts left after removing the development and pilot records. Medium alone gets 9.9% of them right and small gets 8.3%, and 780 facts are right in medium and wrong in small. **Full replacement** of small's layer-8 state by medium's translated state rescues 262 of those 780 (33.6%, with controls at most 0.8%). But it breaks 22% of the facts both models knew and 70% of those only small knew, so small's accuracy falls from 8.30% to 7.17%. The pre-set bar (50% rescue, accuracy half-way to medium, and 90% of known facts kept) was not met, although the controls show that real information does cross.

We then tried a **softer mix**, with its rule written before we built it. We chose $\gamma \in \{0.1, \ldots, 1\}$ on one half of the facts and report the other half (10,895 facts). Mixing in half of medium's state ($\gamma = 0.5$) raises small's accuracy from 8.5% to **9.5%**, where medium alone gets 10.2%. It gains 162 facts and loses 53 ($p = 2.5 \times 10^{-14}$), keeps 98% of the facts both models knew and rescues 28.6% of the facts small gets wrong (Figure 5.3). This meets the rule (a gain of at least 0.5 points, 90% of known facts kept, $p < 0.01$) and is our first net-positive Import result. It is only a one-point gain, it needs medium to run on the same sentence, and its controls were not rerun.

5.7 What the Side Study Shows

A linear channel between GPT-2 Small and Medium exists and keeps the receiving model working. The change an edit makes in small moves medium's answers in a specific, graded way and without damaging its neighbours. But every translator that does not look at medium's output stops at about 20% top-1, far below the 50% bar, while looking at medium for every sentence reaches 95%. Together with the controls, this puts the limit on the *edit's compatibility* and not on the channel, the layer, the translator's capacity or the amount of data. What crosses

is a push toward the targeted word and not a fact. The map follows earlier stitching work [40], and the limits of the study are listed in Chapter 7.

Chapter 6

Implementation

6.1 Software, Hardware and Application

The system is written in Python with PyTorch [20], TransformerLens 3.5.1 [13] (which gives every GPT-2 activation a named hook point and can run the model from a saved intermediate state), SAELens 6.45.3 [14] (which loads the `gpt2-small-res-jb` dictionaries [15]), Neuronpedia [18] for feature descriptions, and Streamlit for the application. The main-line studies ran on a GTX 1660 Ti PC (PyTorch 2.6.0, CUDA 12.4). Caching, the headroom study, the strength models and the transfer study ran on an RTX 4050 laptop (PyTorch 2.5.1, CUDA 12.1).

The application `experiment_app.py` has seven tabs and three tiers: the interface, the intervention engine, and the model and data layer (cached GPT-2, SAEs, fact sets and per-prompt caches). The *Hybrid mute and boost* tab carries both editors behind a choice of editing method. You can pick the fixed sweep, with its strengths, cumulative sweep, pool refill, safety filter and GPU-batched switch, or gradient descent, with its steps, learning rate, margin and weights. It also has a live tracker, a stopwatch and a session-history record. The *Prototype lab* runs gradient descent with the optional additive edit, shows the baseline and edited text side by side, and applies the same edit to follow-up prompts. *Batch* runs studies in background workers, *Repair & SAE limit* shows the diagnostics of Section 4.7, *Sequential vs Batched Proof* times the two code paths, *Monosemanticity Analysis* inspects features, and *Session history* stores every run. The edit itself lives in `src/hooks.py` (Equation 3.2), `src/editing.py` (screening, causal ranking and the safety checks of Equation 3.4), `src/hybrid_runner.py` (the headless sweep, which reproduces the application exactly) and `src/gradient_editing.py` (Equations 3.5 to 3.9). The transfer study adds only new files (`src/transfer/`, and tools `00` to `27` in `tools/transfer/`). We import older functions and never edit them, so earlier results cannot change.

6.2 Correctness Checks and Engineering Lessons

We checked every fast path against a slower reference before using it, and we kept the checks as scripts. Batched ranking matched sequential ranking (identical feature lists, probabilities to 10^{-9}, and 60 of 60 safety decisions). The cached-state path matched the full model (identical logits on one machine, within 10^{-5} on the other). A random per-feature edit through the cache path gave the same target rank as the real hook, and the gradient-descent tuning path agreed with the real hook in every recorded run. Built-in sanity checks stop a run if something does not match, and one of them caught an error in the cache builder (it passed a multiplier of 1.0 where 0.0 was meant) before any data was written. Three problems could have ruined results without raising any error.

- **A thread race removed the start-of-text token.** In TransformerLens, tokenising with the start token switched off temporarily changes a setting shared by the whole model. When two Streamlit sessions overlapped, the wrong value could be written back for good, and every later prompt was tokenised without its start token (rank 28 instead of 22 on one prompt). Of 57 recorded application runs, 16 had wrong baselines, all from one server process, and none of them is used in a table. The fix tokenises directly and never touches the model's configuration.
- **A timer measured the network** (46 s against about 10 s of model time), and **chained hooks disagreed with the combined check** (rank 66 against 59). We moved the look-ups out of the timed region and now use one combined hook.
- **GPU memory.** A padded training set of 7,550 texts (3.8 GB) nearly filled the 6 GB card and made runs five to ten times slower with no error at all (6.2 hours instead of one). Training tensors now stay on the CPU.

6.3 Reproducibility and Development Process

Every tool prints its pass rule, saves its raw output as a log, and writes a result file with the per-run details. The logs and results of every experiment are copied into a data pack under version control, and the figures in this report are drawn from that pack by scripts that run no model. We optimised in one bounded step at a time: find one bottleneck, measure the reference, make one change, check equivalence on a fixed case, log it, and only then approve the next step.

The software and this report were developed with AI coding assistants (Antigravity AI earlier, Claude later) under the authors' direction. The authors chose the questions, pass bars, controls and each next experiment, and ran or approved every long job. The assistants wrote and tested code, drafted documentation, and were asked to record failures as carefully as successes. [TODO: check this statement against the college's policy; delete it if disclosure is not required]

Chapter 7

Discussion, Conclusion and Future Work

7.1 What the Work Shows

The sweep was held back by the supply of features, not by the selection rule. Taking candidates from all prompt positions made the pool about five times larger and raised the prompts at rank 1 from 4 to 10 of 13. Choosing the right features and strengths mattered more than stronger interventions. Muting the competitor's drivers and adding boosts gave the early gains. One better global strength pair lifted a stand-in success rate from 25% to 39%, and choosing strengths per prompt added nothing we could measure. Tuning one multiplier per feature on the prompt itself reached 271 of 300 on the real model.

Rank 1 turned out to be a weak test. The additive edit forced random words to rank 1 on 20 of 20 prompts, and an optimiser with enough freedom can reach any output you pick. Being able to steer a target to the top says nothing about what the model knows, in either direction. The edit itself is a push toward the answer's output direction. It carries little beyond the prompt it was tuned on, and it mostly ignores an instruction not to say the word. The side study showed that a linear channel between GPT-2 Small and Medium keeps the receiving model working, that an exported edit moves the target in a specific way but rarely all the way to rank 1, that what crosses is a push and not a fact, and that mixing in half of Medium's state raises Small's accuracy by one point.

Four habits paid off: controls first, fair comparisons, pass bars fixed before each run, and corrections made in the open. Withdrawn claims are marked in the journal where they happened.

7.2 Threats to Validity and What Is Not Shown

- We use one model family, one older SAE release, and layer 8 in all later studies. The early studies use small, templated prompt sets (13 to 131 prompts), and like CounterFact they treat prompts as independent, without correction for multiple comparisons.

- Success means rank 1 on the next token, and the median target probability at rank 1 is only 7.7% for gradient descent. We measured side effects on the same prompt, on 20 unrelated prompts and in 40-prompt tests, with one seed, no confidence intervals, and true targets only in the generation and instruction tests.
- The headroom and the learned strengths are stand-ins measured on cached states, and we never ran the real sweep with the learned strengths. The side study has 88 primary test records.
- **We have not compared the approach with ROME, MEMIT, in-context editing, prompting or difference-of-means steering**, so nothing here says how it ranks against them.

7.3 Conclusion

FeatureScalpel set out to repair suppressed factual recall in GPT-2 Small through the features of a sparse autoencoder, without changing a single weight. After many attempts that mostly failed in useful ways, we arrived at a transient edit that brings the true answer to rank 1 on 271 of 300 held-out prompts, quickly and with small side effects on the same prompt. The applied side study found that the change an edit makes can be carried between models as a push, that importing a larger model's state gives a small gain, and that a fact does not cross. The results support the claim that a transient SAE-feature edit can raise a suppressed answer to the top, and that tuning a multiplier per feature does so more often, faster and with lower same-prompt side effects than a hand-built sweep. They do not support the claim that such edits recover knowledge, that they are specific to a fact, or that they beat weight editing, in-context editing or prompting.

7.4 Future Work

1. **A specificity benchmark.** For every method, on the same prompts, it should measure whether the target reaches rank 1, carries over to rewordings, leaves unrelated prompts and generated text alone, and respects an instruction, with random words of matched starting rank as controls.
2. **A head-to-head comparison** with ROME, MEMIT, in-context editing and difference-of-means steering, using existing implementations [30] and timing everything on one machine.
3. **More strengths, transfer and scale.** We would run the learned strengths in the real sweep, rerun the controls for the Import mix, and try other model pairs (for example

Gemma 3 270M and 1B, which have SAEs on both sides) and larger models with pre-trained SAEs [17].

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Appendices

Appendix A: Research Journal and Report Sections

The journal (`docs/Research_Journal/`, entries 1 to 37) records every step with its tables, rules, mistakes and corrections. Result files and raw logs of the later entries are in `docs/Research_Journal/packs/`.

Table A.1: Journal entries and where they appear in this report.

Entries	Content	Section
1	First prototype, causal selector, eligibility rule, roadmap	4.1
2, 3	Target-helper and competitor ablation (1 and 3 of 22)	4.2
4, 6, 7, 8	Weighted multi-competitor muting, aggregation, decay profile, top-K	4.3
5	Hybrid mute and boost, rank regression	4.4
9, 10, 11	SAE audit, layer benchmark, caching and profiling	4.5
12, 13, 14, 15	GPU, redundant-pass removal, batching (186 to 6 calls), batched application	4.5
16, 17, 18, 19	Best-so-far, cumulative sweep, combined hook, pool refill, all-position candidates	4.6
20, 21	Candidate-source study (10 of 13 against 4 of 13); safety-filter study	4.6
22	Repair and SAE-limit diagnostics	4.7
23, 24	Learned-strength plan; CounterFact hard set, split, cache, headroom	4.8
25	Literature check	4.7, Ch. 2
26, 27	Learned strengths (PromptNet, FeatureNet)	4.9
28, 29	Gradient-descent editor; equal-budget comparison (271, 161, 94 of 300)	4.10
30	Additive edit, random-word control, rewording, generation and instruction tests, start-token bug	4.11, 4.12
31 to 37	Cross-model transfer: translators, country swap, Export, search ladder, Import, SAE control	Ch. 5

Appendix B: Pass Bars of the Side Study and Their Verdicts

Every bar was written into the project plan, and committed, before the run that judged it; none was moved afterwards. "Edit's own" means the effect that the source edit has inside small itself. Verdicts are as printed by the tools.

Table B.1: Pass bars ("gates") and their verdicts.

Step	Bar fixed before the run	Result	Verdict
Gate 1 (A2)	Translator: difference cosine at the last position at least 0.3	Import pairs 0.31; Export pair 0.28	Import only
Gate 2 (B1)	Country swap beats the strongest control by a margin, paired test	55% and 33% against at most 0.4%	pass
Gate 3 (D)	Export top-1 at least 15% and 10 points above the strongest control	8%; margin 3 points	fail
Gate 4 (D5)	Best layer set at least 10 points above the best single layer, $p < 0.01$, top-1 at least 15%	7% against 8%; $p = 0.016$	fail
Gate 5 (D6)	Top-1 $\geq$ 50%; rewordings lift $\geq$ 80% of the edit's own; neighbours fall $\leq$ 5 points	95%; 0.91; 8 points	fail (neighbours)
Gate 6 (D7)	Top-1 $\geq$ 50%; rewordings lift $\geq$ 0.8; neighbours fall no more than the edit's own	18%; 0.74; 7 points	fail
Gate 7 (D10)	As Gate 6, plus real target 10 points above the random word	20%; 0.74; met; met	fail
Gate 8 (Import)	Rescue $\geq$ 50% of the 780 facts; accuracy halfway to medium; 10 points above controls ($p < 0.01$); 90% of known facts kept	33.6%; 7.2% against 9.1%; met; 78.3%	fail
Gate 9 (D13)	Gate 7 unchanged, on the full varied training set	14%; 0.62; met; met	fail
Rule E5	Soft Import: gain $\geq$ 0.5 points on held-out facts, 90% kept, $p < 0.01$	+1.0 point; 98.3%; $p = 2.5 \times 10^{-14}$	met
Rule S1	SAE basis "matters" if top-1 is higher by 10 points or rewordings lift by 5	+15 points; +2 points	matters
Rules S2, S3	Real target's rewordings lift larger than a random word's, $p < 0.01$	$p = 0.031$ (D6); $p = 0.36$ (D7)	not shown

Two notes written after the fact do not change any verdict: the 5-point neighbour line of Gate 5 was stricter than the source edit's own loss of 9 points inside small, so Gates 6, 7 and 9 use the edit's own loss; and the dose rule (dev top-1 on 28 records) was replaced by the dev log-rank gain from step D10 on, after step D9b showed it to be fragile.

Appendix C: Reproducing the Results

From the repository root, with `HF_HUB_OFFLINE=1` once the models are cached. The main line is on the branches `layer-intervention-benchmark`, `pool-refill-implementation` and `gradient-descent-editing`; the side study is on `cross-model-transfer`.

```
streamlit run experiment_app.py
python run_batch.py data/candidate_source_batch_spec.json
python tools/build_counterfact_set.py
python tools/split_counterfact_set.py
python tools/build_strength_cache.py
python tools/measure_headroom.py
python tools/train_strength_models.py
```

```
python tools/compare_sweep_vs_gradient.py --n-per-band 0
python tools/control_random_targets.py
python tools/transfer/00_setup_check.py   (then 01 to 27)
python tools/make_report_v3_history_figures.py
```

In order: the application; the candidate-source study; the CounterFact hard set and split; the per-prompt cache and the headroom measurement; the strength networks; the 271, 161 and 94 of 300 comparison; the random-word control; the side study; and the early-stage figures.

Journal entries 31 to 37 give each side-study tool with its run time, and `docs/cross_model_transfer/PLAN.md` holds all pass bars and readings.